

Wireless Standards — Guided Notes ANSWER KEY

Unit 2.2.1 | CompTIA Network+ N10-009 Obj. 2.2 | N10-008 2.2

For instructor use / distribute after students complete the notes.

Question 1

The 2.4 GHz band offers better _____ and _____ than 5 GHz but has only _____ non-overlapping channels in the US and suffers from more _____ from non-Wi-Fi devices.

Answer: Range; wall penetration; three (channels 1, 6, 11); interference.

Question 2

The 6 GHz band, used by Wi-Fi _____ and Wi-Fi _____, provides approximately _____ MHz of spectrum. Its main limitation compared to 5 GHz is _____ and _____.

Answer: Wi-Fi 6E; Wi-Fi 7; 1200 MHz; shorter range; worse wall penetration.

Question 3

Fill in the theoretical maximum throughput for each standard:

802.11b → _____

802.11a → _____

802.11g → _____

802.11n → _____

802.11ac → _____

802.11ax → _____

Answer: b: 11 Mbps; a: 54 Mbps; g: 54 Mbps; n: 600 Mbps; ac: 6.93 Gbps; ax: 9.6 Gbps.

Question 4

MIMO uses multiple _____ to send parallel _____ over the same channel. A '4x4' device has _____ transmit and _____ receive antennas. If a 4x4 AP talks to a 2x2 client, the client gets _____ streams.

Answer: Antennas; spatial streams; 4; 4; two streams (limited by the client).

Question 5

MU-MIMO allows the AP to communicate with _____ clients simultaneously. Wi-Fi 5 added MU-MIMO on the _____. Wi-Fi 6 added MU-MIMO on the _____ as well.

Answer: Multiple; downlink (AP to clients); uplink (clients to AP).

Question 6

Wi-Fi 6 introduced _____ which lets the AP divide a channel into sub-channels to serve multiple clients per transmission. _____ reduces interference by tagging frames so devices ignore signals from neighboring networks on the same channel.

Answer: OFDMA (Orthogonal Frequency Division Multiple Access); BSS Coloring.

Question 7

In a high-density environment, wider channels provide _____ per client but leave _____ non-overlapping channels. Best practice for dense deployments is _____ channels with _____ AP density.

Answer: More throughput; fewer; narrower; higher.

Question 8 — Real World Application

REAL WORLD: A school deploys Wi-Fi 6 APs in every classroom. During standardized testing, 30 Chromebooks per room are all active simultaneously and performance is poor. What Wi-Fi 6 features specifically address this scenario, and what configuration changes would you make?

Answer: OFDMA addresses high-density small-packet traffic by serving multiple clients per transmission instead of taking turns. MU-MIMO (uplink and downlink) allows simultaneous transmissions. BSS Coloring reduces co-channel interference from adjacent classroom APs. Configuration: ensure 5 GHz with 20 MHz channels (maximizes non-overlapping channels), verify AP placement gives sufficient coverage without excessive overlap, enable band steering to push capable devices to 5 GHz, and confirm OFDMA and MU-MIMO are enabled in the AP configuration.